

Business Summary Report:

Predictive Insights for Collections Strategy

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1. Summary of Predictive Insights

Our predictive model analyzed customer payment history, credit behavior, and demographic data to identify the strongest indicators of delinquency risk. Three consistent patterns emerged across the dataset:

- **High Credit Utilization:** Customers utilizing over 80% of their available credit are significantly more likely to become delinquent, regardless of income level.
- **Recent Payment Behavior:** A pattern of late or missed payments in the preceding 2–3 months is the single strongest early-warning signal for future default.
- **Elevated Debt-to-Income Ratio:** Customers with a Debt-to-Income Ratio above typical thresholds, particularly self-employed and variable-income segments, show elevated risk even when total income appears sufficient.

Key Insight Summary Table

Key Insight	Customer Segment	Influencing Variables	Potential Impact
High utilization drives risk	Revolving credit users, mid-to-high balances	Credit_Utilization, Credit_Score	Early flag for proactive outreach before missed payment occurs
Recent missed payments predict default	Customers with 1+ missed payment in last 3 months	Month_1-Month_6 payment history	Prioritize this group for immediate collections contact
High debt load relative to income	Self-employed / variable-income customers	Debt_to_Income_Ratio, Employment_Status	Tailor repayment plans instead of standard collections track

2. Recommendation Framework

Restated Insight

Customers with credit utilization above 80% show a sharply increased probability of delinquency, even before a missed payment appears on record.

Proposed Recommendation

- **Specific:** Launch a proactive early-intervention outreach program for customers whose credit utilization exceeds 80%.
- **Measurable:** Target the top 20% highest-risk customers by model-generated risk score each month.
- **Actionable:** Trigger automated alerts to the Collections team and send personalized repayment or credit-counseling offers before delinquency occurs.
- **Relevant:** Directly supports Geldium's goal of reducing default rates and collections costs through earlier, lower-friction engagement.
- **Time-bound:** Pilot the program over one quarter, with a review of delinquency-rate impact at the 90-day mark.

Justification and Business Rationale

Acting on the high-utilization signal before a payment is missed shifts Collections from a reactive to a preventive model. Since utilization is already a top-5 predictive feature, this approach reduces manual review effort while directing outreach to the customers most likely to benefit from early support.

3. Ethical and Responsible AI Considerations

- **Fairness Risk:** The model could disproportionately flag lower-income or self-employed customers, who may naturally carry higher utilization. Mitigation: monitor the Disparate Impact Ratio (target 0.80–1.25) across income and employment segments, and adjust thresholds if bias is detected.
- **Explainability:** A complex model can be hard for non-technical stakeholders to trust. Mitigation: use SHAP explainability outputs to communicate, in plain terms, which specific factors (e.g., "high utilization" or "recent missed payment") drove each customer's risk score.

This approach supports responsible AI use by keeping a human-in-the-loop for final collections decisions, maintaining transparency on which data drives predictions, and avoiding automated action that could penalize customers based on demographic factors rather than genuine payment risk.